AP CALCULUS AB

Unit 1 • Lesson 1.4 practice • Answer key

Lesson 1.4 • Estimating Limit Values from Tables

5 multiple-choice questions and 1 free-response question • Answer key with explanations

Directions

Questions 1 to 5 are multiple choice: choose the best of the four answer choices. Question 6 is free response: show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

x	1.9	1.99	1.999	2.001	2.01	2.1
$f(x)$	0.47	0.497	0.4997	0.5003	0.503	0.53

Which is the best estimate of $\lim_{x \to 2} f(x)$?

(A) 0.5

(B) 0.4997

(C) 2

(D) It does not exist.

Solution

Answer: (A). The values approach 0.5 from both sides.

Question 2

NO CALCULATOR

As $x \to 3^-$, the values of $g(x)$ are 1.9, 1.99, 1.999; as $x \to 3^+$, they are -0.9, -0.99, -0.999. What is $\lim_{x \to 3} g(x)$?

(A) 2

(B) -1

(C) 1

(D) It does not exist.

Solution

Answer: (D). The left-hand limit appears to be 2 and the right-hand limit -1.

Question 3

NO CALCULATOR

As $x \to 0$ from both sides, $h(x)$ takes the values 10, 100, 1000, 10000. Which statement is true?

(A) $\lim_{x \to 0} h(x) = 10000$

(B) h increases without bound near 0; the limit does not exist as a real number.

(C) $\lim_{x \to 0} h(x) = 0$

(D) $h(0) = 10000$

Solution
Answer: (B). The outputs grow without bound: $\lim_{x\to 0} h(x) = \infty$.

Question 4

NO CALCULATOR

Which is true about using a table to find a limit?

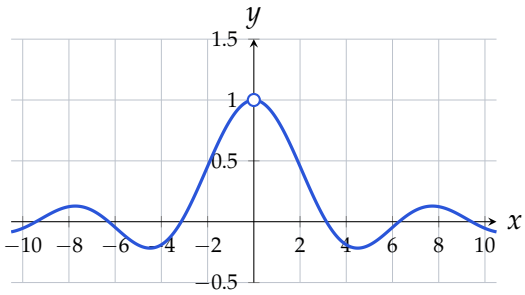
(A) A table always gives the exact limit.
(B) A table cannot suggest a limit.
(C) A table suggests a value but does not prove the limit.
(D) A table needs only values on one side.

Solution
Answer: (C). Tables give evidence; algebra or theorems confirm the value.

Question 5 (graph)

NO CALCULATOR

The graph of $y = \dfrac{\sin x}{x}$ is shown; it has a hole at $x = 0$. Which is the best estimate of $\lim\limits_{x\to 0} \dfrac{\sin x}{x}$?



(A) 0
(B) π
(C) 1
(D) It does not exist.

Solution
Answer: (C). The graph approaches the open circle at height 1 from both sides, so the limit appears to be 1 (this is proved with the squeeze theorem in Topic 1.8).

Question 6 (Free response)

NO CALCULATOR

x	-0.1	-0.01	-0.001	0.001	0.01	0.1
$f(x) = \frac{e^x-1}{x}$	0.9516	0.9950	0.9995	1.0005	1.0050	1.0517

(a) Use the table to estimate $\lim\limits_{x\to 0} f(x)$.
(b) Is $f(0)$ defined? Does this affect your answer to (a)?
(c) A function g has $g(1.9) = 1.9$, $g(1.99) = 1.99$, $g(1.999) = 1.999$, $g(2.1) = 3.1$, $g(2.01) = 3.01$, $g(2.001) = 3.001$. Estimate the one-sided limits of g at $x = 2$. Does $\lim\limits_{x\to 2} g(x)$ appear to exist?

Solution
(a) The values approach 1 from both sides, so the limit appears to be 1.
(b) No, $f(0) = \frac{0}{0}$ is undefined. This does not affect the limit, which depends only on values near 0.

(c) $\lim_{x \to 2^-} g(x)$ appears to be 2 and $\lim_{x \to 2^+} g(x)$ appears to be 3. They differ, so $\lim_{x \to 2} g(x)$ does not appear to exist.